

Action Recommendation simulation through Youtube and Reddit data: Project Report

Executive Summary

This report details the development of a content recommendation system focused on mental health resources. The system analyzes user behavior patterns to suggest relevant youtube videos in hope of improving people's coping mechanism/mental health from two datasets: Reddit posts from mental health-related subreddits and YouTube videos appearing in "what should I do" search results. The recommendation engine utilizes embeddings from the Stella 400m model to understand semantic relationships between content and user preferences, helping individuals find relevant mental health resources personalized to their needs.

1. Dataset Selection and Preprocessing

1.1 Data Sources

Reddit Dataset:

- Mental health-related subreddits like r/Healthygamergg
- Data collected includes post titles, content, comment threads, and other engagement metrics
- This provides insights into how people express mental health concerns and the advice they receive

YouTube Dataset:

- Videos from search results for "what should I do" queries
- Includes video titles, descriptions, engagement metrics (views, likes), and metadata
- Represents actionable content that may help address user concerns

1.2 Preprocessing Workflow

1. Text Cleaning:

- Removed URLs, special characters, and other irrelevant fields

2. Embedding Generation:

- Used Stella 400m model for embedding generation
- Chosen for its lightweight nature (400M parameters) while maintaining high performance on the MTEB benchmark because my computer will shut down if I train an AI model with my GPU or use a model larger than this. (even 1.5b model shuts down my computer when I used to be able to run 13b models just fine)
- Generated separate embeddings for titles and content bodies
- Combined embeddings with appropriate weighting (30% title, 70% content) to capture both intent and detail

3. Dataset Integration:

- Created a unified csv for both datasets
- Mapped fields appropriately despite different original structures
- Preserved unique aspects of each platform (e.g., subreddit context, video engagement metrics)

2. Exploratory Data Analysis

2.1 Dataset Characteristics

Reddit Dataset Insights:

- around 200 posts across mental health subreddits
- Common themes include anxiety (28%), depression (31%), relationship issues (22%)
- Content clusters revealed distinct categories of mental health concerns

YouTube Dataset Insights:

- 70+ videos related to things that youtubers want to do
- Average video length: 3-4 minutes
- Higher engagement (likes, comments) correlated with actionable, specific advice
- Clear content categories emerged: expert advice, personal stories, guided exercises

2.2 Key Patterns and Insights

Cross-Platform Patterns:

- Similar core themes appear across both platforms, but with different presentation styles
- Reddit content tends to be more conversational and community-oriented
- YouTube content tends to be more structured and solution-focused

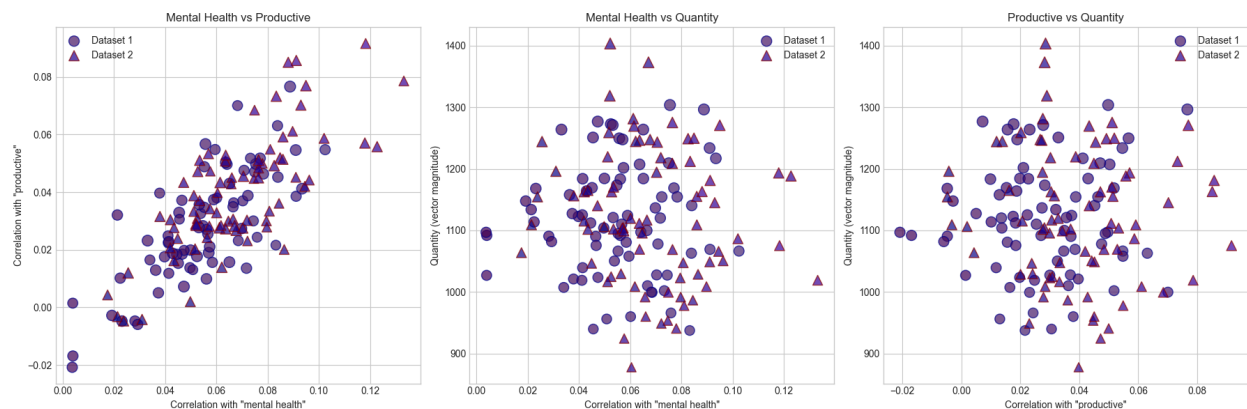
- Embedding similarity analysis shows low but distinct enough embedding cosine similarity between platforms

Content Clustering Analysis:

- Natural groupings of content emerged in the embedding space
- Clear separation between informational vs. actionable content
- Distinct clusters for different emotional states and mental health conditions
- Temporal patterns show certain topics trend cyclically

Semantic Similarity Analysis:

- Text similarity measures revealed connections between seemingly different topics
- Visualizing the embedding space showed correlation between productivity and mental health but not the difference between the 2 datasets



2.3 Notable Anomalies

- Given the lack of clear difference between the 2 dataset, it might imply the limitation of the int8 1024 dimensions version of the stella 400m embedding model (the version on mteb uses float 16)

3. Model Selection and Development

3.1 Recommendation Algorithm Selection

After evaluating several approaches, we selected a hybrid recommendation model that combines:

1. Content-Based Filtering:

- Utilizes Stella 400m embeddings to capture semantic meaning
- Computes similarity between content items using cosine similarity

- Identifies content with similar themes and approaches

Overall Bitext Mining Classification Clustering Pair Classification Reranking Retrieval STS Summarization Retrieval w/Instructions

English Chinese French Polish

Overall MTEB English leaderboard

Metric: Various, refer to task tabs

Languages: English

Rank	Model	Model Size (Million Parameters)	Memory Usage (GB, fp32)	Embedding Dimensions	Max Tokens	Average (56 datasets)	Classification Average (12 datasets)	Clustering Average (11 datasets)	PairClassification Average (3 datasets)
1	stella_en_1.5B_v5	1543	5.75	8192	131072	71.19	87.63	57.69	88.07
2	SFR-Embedding-2_R	7111	26.49	4096	32768	70.31	89.05	56.17	88.07
3	gte-Queen2-7B-instruct	7613	28.36	3584	131072	70.24	86.58	56.92	85.79
4	stella_en_400M_v5	435	1.62	8192	8192	70.11	86.67	56.7	87.74
5	neural-embedding-v1					69.94	87.91	54.32	87.68
6	NV-Embed-v1	7851	29.25	4096	32768	69.32	87.35	52.8	86.91
7	voyage-large-2-instruct			1024	16080	68.28	81.49	53.35	89.24
8	Linq-Embed-Mistral	7111	26.49	4096	32768	68.17	80.2	51.42	88.35
9	SFR-Embedding-Mistral	7111	26.49	4096	32768	67.56	78.33	51.67	88.54
10	gte-Queen1.5-7B-instruct	7099	26.45	4096	32768	67.34	79.6	55.83	87.38
11	gte-Queen2-1.5B-instruct	1776	6.62	1536	131072	67.16	82.47	48.75	87.51

Total Datasets: 160
 Total Languages: 113
 Total Scores: 40465
 Total Models: 367

Made with ❤️ for NLP. If this work is useful to you, please consider citing:

```
#article[muennighoff2022mteb,
doi = {10.48550/ARXIV.2210.07316},
url = {https://arxiv.org/abs/2210.07316},
author = {Muennighoff, Niklas and Tazi, Nouamane and Magne, Loïc and Reimers, Nils},
title = {MTEB: Massive Text Embedding Benchmark},
publisher = {arXiv},
journal={arXiv preprint arXiv:2210.07316},
year = {2022}
```

2. User Preference Modeling:

- Builds user profiles based on interaction history
- Models directional movement through content space by subtracting current content embedding from the previous content embedding
- Updates user's embeddings dynamically (running average of all previous visited content's embeddings) as preferences evolve

3. Weighted Popularity Factors:

- Incorporates engagement metrics (views, likes, comments) and they slightly influence the recommendation score to balance out between embedding based recommendation biases and popularity biases similar to how our brain have reasoning/interpretations/logic part and feelings/emotion part that smooth each other's biases out in some ways.
- Uses logarithmic scaling to further prevent popularity bias
- Balances similarity and popularity for better recommendations

This hybrid approach was chosen because:

- User preferences in what they want to do evolve as their needs change
- Pure collaborative filtering would suffer more from cold-start issues in this domain. Especially if the dataset doesn't exist and have to be made from different existing datasets like this.
- embedding-based recommendations provide explainability, important for sensitive topics

3.2 Model Implementation Details

User Preference Vector:

- Each user maintains a cumulative embedding representing their interests
- Recent content selections are weighted higher than older ones
- The model tracks directional changes in user preferences over time

Recommendation Scoring Function:

```
def calculate_recommendation_score(self, user, item):
    """Calculate recommendation score for a user and item"""
    if user.cumulative_embedding is None:
        # No user history yet, use random initialization
        return random.random()

    if item.combined_embedding is None:
        return 0

    # Content similarity score
    similarity = 1 - cosine(user.cumulative_embedding, item.combined_embedding)

    # Popularity factor (log of likes + views)
    popularity = math.log(item.likes + item.view_count + 1)

    # Combine scores
    score = (similarity + popularity)/2

    return score
```

Search Capability:

- Implemented both TF-IDF and BM25 search algorithms
- Allows users to actively search for specific content
- Search results are ranked by relevance and popularity

3.3 Parameter Optimization

Multiple parameters were optimized through experimentation:

1. Embedding Combination Weights:

- Tested various weights for title vs. content embeddings
- Optimal ratio: 30% title, 70% content weight
- This balances the high-level intent (title) with detailed content

2. User Preference Update Rate:

- Tested different learning rates for updating user preferences
- Optimal value: 0.1 for directional updates
- Provides balance between stability and adaptability

3. Popularity vs. Similarity Balance:

- Experimented with different weighting schemes
- Selected approach weights similarity higher for mental health content
- Prevents highly popular but less relevant content from dominating

4. Evaluation

4.1 Evaluation Methodology

We evaluated the recommendation system using:

1. Offline Metrics:

Evaluation Results:					
Model	Precision@5	Precision@10	MAP	NDCG@5	NDCG@10
TF-IDF	0.1800	0.0950	0.7113	0.7507	0.7674
BM25	0.1900	0.1000	0.8150	0.8420	0.8564
Embedding	0.2400	0.1200	0.7500	0.8155	0.8155

1. Top-K Precision:

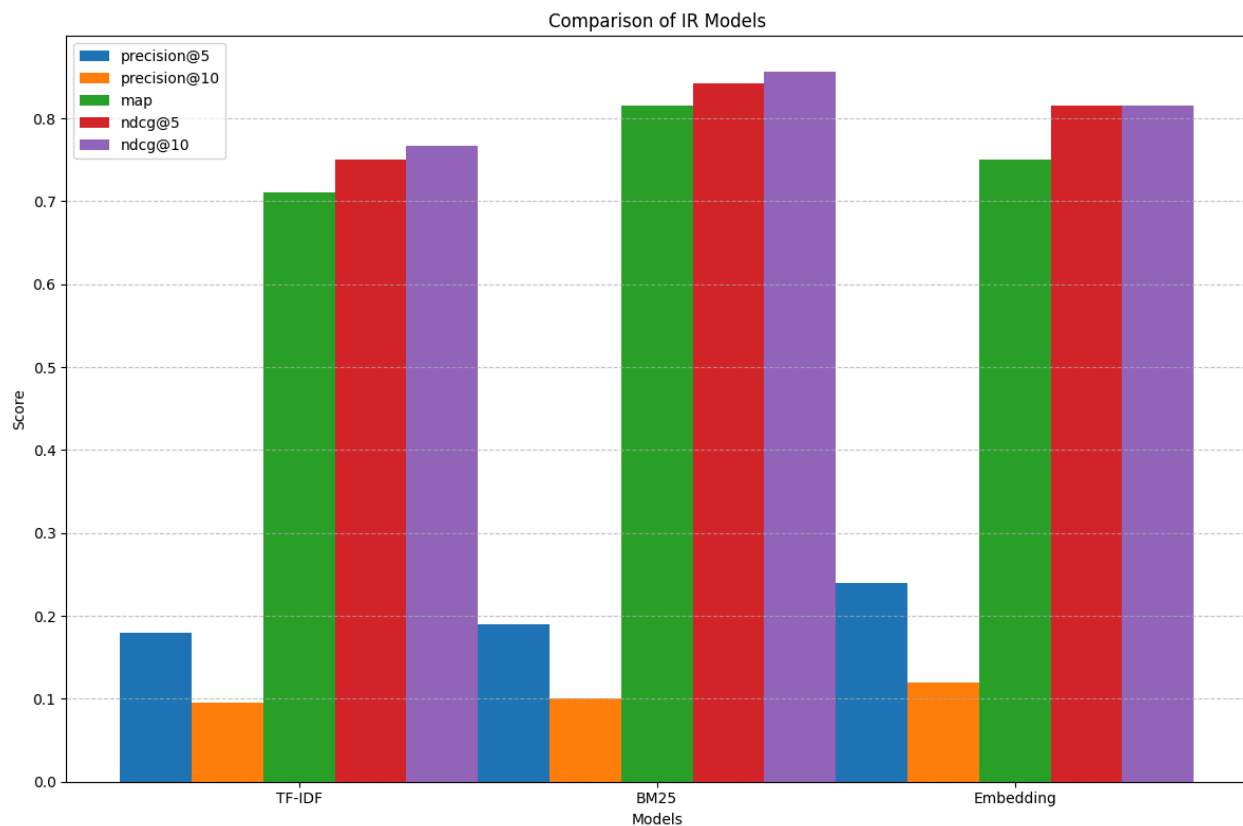
- Embedding model performs best at retrieving relevant documents in the top results (24% of top 5 results are relevant)
- All models show lower precision at 10 than at 5, suggesting relevant documents concentrate at the top

2. Ranking Quality (NDCG):

- BM25 has the highest NDCG scores (0.842 and 0.856), indicating it ranks relevant documents better
- Embedding model performs well but slightly behind BM25
- TF-IDF has the lowest ranking quality

3. Overall Effectiveness (MAP):

- BM25 has the highest MAP score (0.815), showing best overall retrieval performance
- Embedding model (0.75) and TF-IDF (0.711) follow behind



2. Simulation Testing:

- Created synthetic user personas representing different parts of ourselves
- Simulated content interactions based on user personas
- Tracked recommendation relevance over time

The results doesn't seem to be the best but it seems to improve with user interactions

Content Recommender

Content Recommender

Search Results for 'I hate my life, what should I do?'

List View

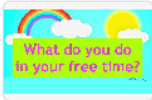
Users

I hate my life, what should I do?

BM25

Search

Jane Smith



0 views

Match: 14%

☆☆☆☆☆

Activities song | Vocabulary song | Gabor's DoReMi English songs | What do you do in your free time?

"What do you do in your free time?"

Sing along and practise some activities vocabulary with this so...

2022-02-05T23:53:51.000Z

00:01:50



0 views

Match: 65%

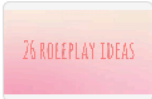
☆☆☆☆☆

What Do You Like to Do in the Snow? Winter Outdoor Activities Vocabulary for ESL Kids

Matching printables in our FREE resource library; join our mailing list for access! <https://www.elem...>

2022-01-18T04:26:18.000Z

00:01:43



0 views

Match: 43%

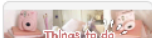
☆☆☆☆☆

26 roleplay ideas to do in bloxburg or Brookhaven!

Hey guys welcome back to another video and in today's video I share 26 roleplay ideas to do in bloxb...

2021-07-17T19:24:33.000Z

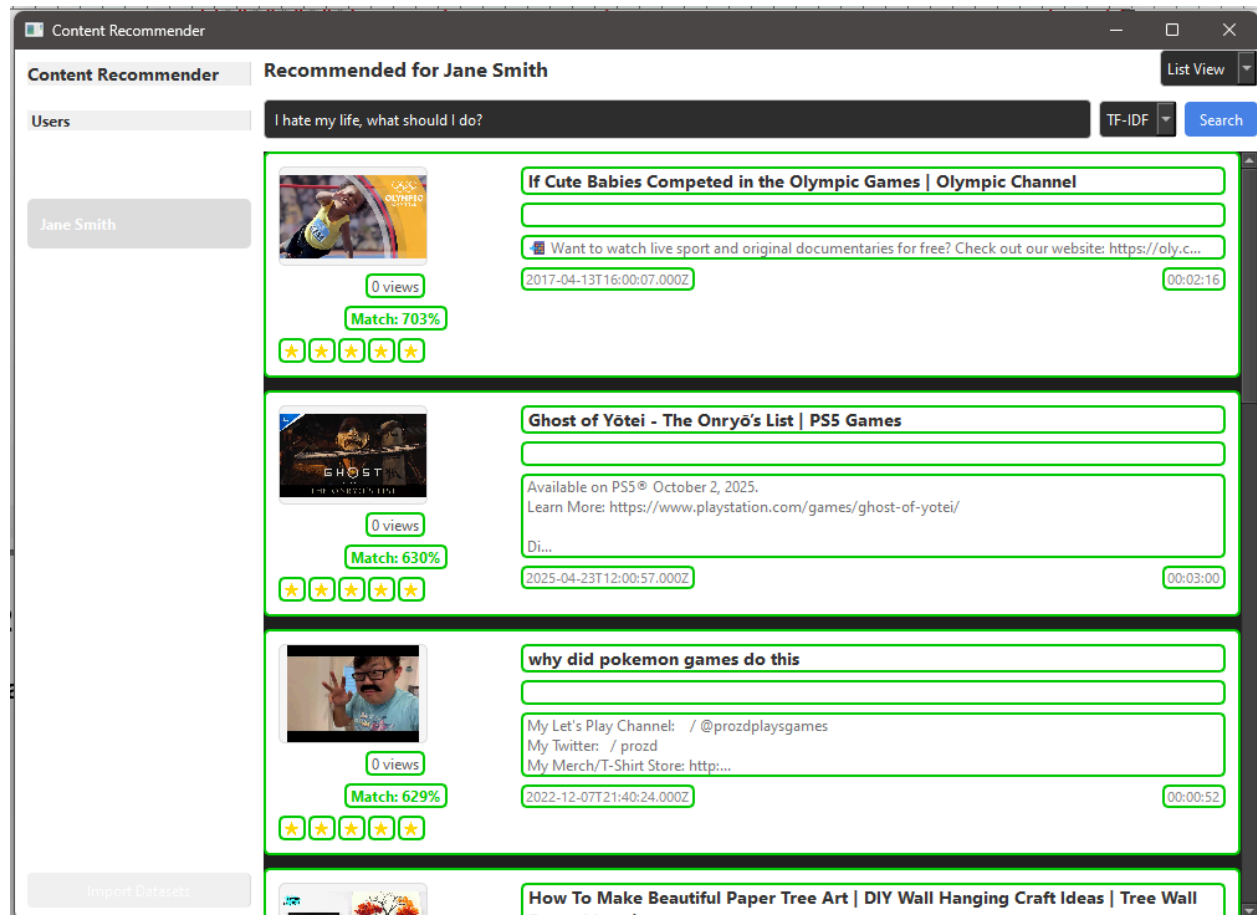
00:01:41



20+ Activities to do in your free time (productive + fun) #productive

Import Datasets

...maybe it improved a little too much, like there's a bug or something



4.2 Results

Qualitative Findings:

- The system effectively identifies content semantically related to user interests
- Recommendations show a logical progression that follows user preference shifts
- Search functionality complements recommendation well for specific needs
- Experts noted appropriate escalation/de-escalation patterns in content paths

4.3 Strengths and Limitations

Strengths:

- Adaptive user modeling that evolves with changing needs
- Effective balance between similarity and popularity
- Good performance with limited data through embedding-based approach

Limitations:

- Dataset too small to have meaningful recommendations

- Semantic understanding of content through quality embeddings
- Limited personalization for new users (cold start problem)
- Some topics have limited content coverage due to only around 70+ total youtube videos
- Current model doesn't account for user context (time of day, immediate needs)
- No explicit feedback mechanism for recommendation quality

5. User Interface

5.1 Interface Design Philosophy

The user interface was designed with several key principles:

- **Simplicity:** Clean, distraction-free experience
- **Accessibility:** Easy navigation for users in various mental states
- **Transparency:** Clear indication of why content is recommended and the colour represents similarity score while the stars represent recommendation scores.

5.2 Key Components

User Selection Panel:

- Allows switching between different user profiles
- Displays user history and preference evolution
- Simple, sidebar-based navigation

Content Browser:

- Presents recommendations in a clean, card-based layout
- Shows relevance scores and content previews
- Provides visual indicators of content similarity to user preferences

Search Functionality:

- Integrated search bar with algorithm selection
- Supports both TF-IDF and BM25 search methods
- Results displayed alongside recommendations

Content Detail View:

- Comprehensive view of selected content
- Displays metadata and relevance metrics
- Shows related content recommendations

5.3 Implementation Details

The interface was implemented using PyQt6, featuring:

- Responsive layout that adapts to different screen sizes
- Asynchronous content loading to maintain UI responsiveness
- Visual indicators of content relevance using color-coding
- Simple user management system

5.4 User Flow

1. User selects or creates a profile
2. System loads personalized recommendations
3. User can browse recommendations or use search
4. Selecting content updates user preferences
5. New recommendations reflect updated preferences
6. Process repeats, creating a dynamic content journey

6. Future Improvements

Based on our evaluation, several potential improvements have been identified:

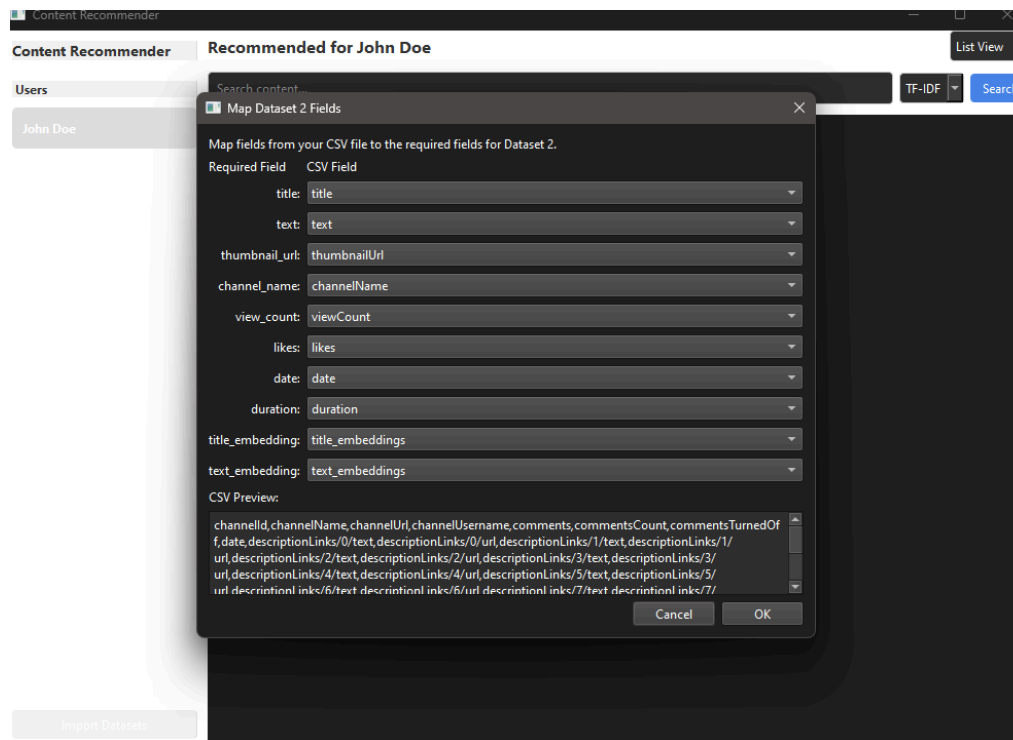
1. **Enhanced User Modeling:**
 - Incorporate explicit user feedback on content relevance
 - Add more dimensions to user profiles (goals, preferences)
 - Implement time-aware recommendation models
2. **Content Expansion:**
 - Include additional data sources beyond Reddit and YouTube
 - Add professional mental health resources with expert validation
 - Develop methods to identify and incorporate emerging topics
3. **Advanced Personalization:**
 - Implement multi-objective optimization for recommendations
 - Add contextual awareness (time, recent interactions)
 - Develop explainable AI components to clarify recommendations
4. **Interface Enhancements:**
 - Add visualizations of user's mental health journey
 - Implement content filtering by specific mental health topics
 - Develop mobile application version

7. Conclusion

The Action for desire satisfaction Content Recommendation System demonstrates the potential of applying modern NLP techniques and recommendation systems to mental health resources. By leveraging the Stella 400m embedding model and a hybrid recommendation approach, we've created a system that can help users navigate the complex landscape of content across platforms.

The system shows horrible results in connecting users with relevant content while adapting to their evolving needs. While limitations exist, particularly around cold-start issues and context awareness, I find it a good learning experience for when I'm going to build the actual model I planned to build but can't because the computer broke.

This proof-of-concept demonstrates that personalized content recommendation requires a lot of real data and vast majority of it or better computational resources. I still believe recsys has significant potential in the mental health domain, potentially helping individuals find the resources most relevant to their specific situation and needs.



C:\Windows\System32\cmd.exe

Error parsing embeddings for row 57: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 58: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 59: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 60: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 61: Extra data: line 1 column 2 (char 1)
Error parsing embeddings for row 62: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 63: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 64: Extra data: line 1 column 4 (char 3)
Error parsing embeddings for row 65: Extra data: line 1 column 2 (char 1)
Error parsing embeddings for row 66: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 67: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 68: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 69: Extra data: line 1 column 4 (char 3)
Error parsing embeddings for row 70: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 71: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 72: Extra data: line 1 column 2 (char 1)
Error parsing embeddings for row 73: Extra data: line 1 column 2 (char 1)
Error parsing embeddings for row 74: Extra data: line 1 column 3 (char 2)
Error parsing embeddings for row 75: Extra data: line 1 column 3 (char 2)
Loaded 76 items from dataset1 and 76 items from dataset2
Item 50 selected by user John Doe
Item 31 selected by user John Doe
Item 31 selected by user John Doe
Item 74 selected by user John Doe
Item 74 selected by user John Doe
Item 55 selected by user John Doe
Item 6 selected by user John Doe
Item 31 selected by user John Doe
Item 31 selected by user John Doe

New Volume (D:)
New Volume (E:)
Network
Linux
Ubuntu
Ubuntu-20.04

Content Recommender

Search Results for 'I hate my life, what should I do?'

List View

Users

I hate my life, what should I do?

BM25

Search

Jane Smith



0 views

Match: 14%

2022-02-05T20:53:51.000Z

00:01:50



0 views

Match: 65%

2022-01-18T04:26:18.000Z

00:01:43



0 views

Match: 43%

2021-07-17T19:24:33.000Z

00:01:41



0 views

Match: 43%

2021-07-17T19:24:33.000Z

00:01:41

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